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Sem-GeoDF: Adaptive Gated Union of Semantic and Geometric Dynamic Feature Rejection for Stereo-Inertial Visual Odometry

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ABSTRACT Dynamic objects break the static-scene assumption used by feature-based stereo-inertial visual odometry (VIO). Semantic segmentation masks can suppress movers but often over-reject useful structure in static scenes, whereas geometry-only epipolar filters need no training labels yet may fail when movers dominate the image. This paper presents **Sem-GeoDF**, an engineering fusion module for stereo-inertial VINS-Fusion that combines a non-blocking YOLO dynamic mask with GeoDF-Adaptive epipolar rejection through an adaptive gated union (logical OR), not intersection. Each branch keeps its own scene gate; candidates are merged under a shared reject-ratio budget; and a three-state online policy (static-safe / dynamic-assist / strong-dynamic) modulates soft masking, hard cull, and adaptive backend residual weights $w_i \in [w_{\min}, 1]$. The policy uses only online signals—burst ratio, strong semantic EMA, and bidirectional semantic–geo overlap on raw GeoDF candidates—and never reads dataset labels, ground truth, or hold-out metrics. Under a fair $1.0\times$ bag-rate protocol with thresholds tuned on VIODE `city_day` only, Sem-GeoDF improves high-dynamic hold-out scenes versus GeoDF-Adaptive (e.g., city_night/3_$high$ -56.9%, parking_lot/3_$high$ -38.2%) while reducing static-scene regression relative to always-on semantic masking (SAD-Sem). We do not claim per-cell superiority over SAD-Sem on mean ATE; the contribution is spectrum robustness, protocol hygiene, and a reproducible four-method evaluation on VIODE ($N = 3$ trials per cell) and EuRoC (static generalization, $N = 1$–2 trials per cell). On the eight dynamic hold-out scenes, Sem-GeoDF is better than GeoDF-Adaptive on 6 and worse on 2 under a $\pm 3\%$ band.

INDEX TERMS Dynamic feature rejection, stereo-inertial odometry, semantic segmentation, sensor fusion, visual-inertial SLAM, residual weighting, YOLO

I. INTRODUCTION

STEREO-inertial visual odometry (VIO) estimates ego-motion by fusing camera tracks with inertial measurements and is a core building block for mobile robots, drones, and AR navigation [1], [2]. Most feature-based front-ends assume that the majority of tracked points lie on a rigid static scene. That assumption fails in traffic-like environments: moving vehicles inject inconsistent image motion into the visual residual and inflate absolute trajectory error (ATE), as stressed by dynamic benchmarks such as VIODE [3].

Two complementary remedy families are widely used. Semantic methods remove features that fall on detector or instance/segmentation masks [4]–[6]. They are effective when the model fires correctly, but false positives can strip static structure, and accuracy becomes coupled to model domain, class taxonomy, and compute budget. Geometric methods reject tracks that violate multi-view rigidity, typically via epipolar Sampson tests and robust fundamental-matrix estimation [7]. They require no semantic training data, yet a fundamental matrix estimated from a mover-dominated image can itself become unreliable.

This paper addresses a practical systems question for stereo-inertial VINS: How can semantic and geometric experts be combined online, without oracle scene labels, so that static safety stays close to geometry-only behavior while high-dynamic accuracy exceeds geometry alone? We propose **Sem-GeoDF**, a policy-gated union (OR) of YOLO and GeoDF-Adaptive candidates under one shared reject-ratio guard, with optional adaptive backend residual weighting for suspicious survivors.

a: Contributions.

1) An online adaptive gated-union front-end for stereo-inertial VINS that ORs semantic and GeoDF dynamic candidates with a shared ratio/min-feature guard and non-blocking mask synchronization.
2) A label-free three-state policy (static-safe / dynamic-assist / strong-dynamic) driven by burst/strong EMA and bidirectional overlap on raw GeoDF candidates, repairing a previously dead overlap trigger.
3) Adaptive backend residual weights w_i with Ceres $\sqrt{w_i}$ scaling, computed only from online semantic/geometric evidence and per-id recovery.
4) A reproducible evaluation: fair $1.0\times$ bag rate, thresholds selected on `city_day` only, hold-out `city_night`/`parking_lot` ($N = 3$) and an EuRoC static check ($N = 1$–2 trials per cell), with a four-method matrix switched by `METHODS`. All reported tables are regenerated directly from run artifacts by one script, with the per-cell trial count printed in every cell.
5) Honest positioning: we emphasize spectrum robustness and static safety rather than claiming universal best ATE against semantic-only baselines.

II. RELATED WORK

A. DYNAMIC VISUAL AND VISUAL-INERTIAL SLAM

Handling dynamic scenes has been a long-standing goal for feature-based SLAM [8]. DynaSLAM and DS-SLAM combine geometric motion cues with semantic segmentation for visual SLAM [4], [5], and DynaSLAM II tightly couples multi-object tracking with the estimator [9]. More recent semantic-geometric systems refine this idea: RDS-SLAM removes the tracking-thread stall of semantic masks [10], CFP-SLAM uses a coarse-to-fine dynamic-probability model [11], Crowd-SLAM targets crowded scenes with detection [12], and DytanVO jointly refines odometry and motion segmentation [13]. On the visual-inertial side, DynaVINS adds robust pose-graph factors for dynamic environments [6], and Liu et al. [14] combine object detection with IMU for resource-restricted RGB-D robots. Those full-stack systems are strong references and generally target either RGB-D or full SLAM back-ends. Our scope is intentionally narrower and complementary: a drop-in stereo-inertial front-end module for VINS-Fusion with explicit realtime constraints (non-blocking masks), an online label-free policy, optional backend residual weighting, and protocol hygiene (no label leakage into the online estimator). We do not claim to outperform these full stacks; we target a reproducible, honestly-scoped fusion component.

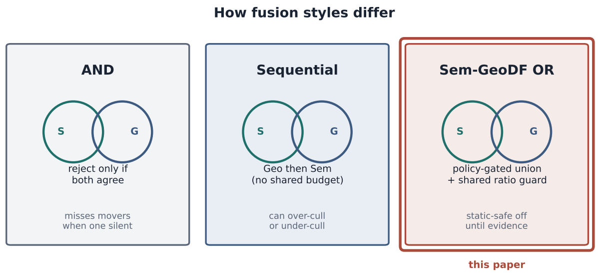

FIGURE 1. Conceptual contrast of fusion styles. Sem-GeoDF uses a policy-gated union (C

TABLE 1. Conceptual design contrast among fusion styles.

Aspect	AND	Sequential	Sem-GeoDF
Reject if either expert fires	no	partial	yes (OR)
Shared reject-ratio budget	rare	no	yes
Static-safe semantic hard-cull off	rare	no	yes
Label-free online policy	varies	varies	yes
Adaptive backend residual weights	rare	rare	yes (optional)

B. GEOMETRY-ONLY DYNAMIC FEATURE REJECTION

Epipolar geometry and Sampson residuals are classical tools [7]. Always-on geometric filters risk removing static structure during aggressive rotation or low parallax. GeoDF-Adaptive introduces scene-aware activation with an auto-calibrated ρ_{on} and track-level voting; Sem-GeoDF reuses this branch unchanged as the geometric expert.

C. SEMANTIC MASKING IN VIO

Real-time instance/segmentation networks such as YOLO [15], [16] provide dynamic-object pixels at interactive rates. Our SAD-Sem ablation applies semantic masking alone under the same fair bag rate and therefore isolates the benefit of fusion versus "semantics always on."

D. HOW SEM-GEODF DIFFERS

Table 1 and Fig. 1 contrast design choices. AND fusion rejects only when both experts agree and can miss movers when one expert is silent. Sequential cascades apply one filter after the other without a shared budget. Sem-GeoDF uses policy-gated OR with a shared reject-ratio guard and a static-safe mode that disables semantic hard cull until online evidence escalates.

III. METHOD

A. SYSTEM OVERVIEW

Fig. 2 gives a full-system view of Sem-GeoDF inside stereo-inertial VINS-Fusion; Fig. 3 compresses the same front-end into a per-frame flow. Camera images feed YOLO asynchronously (`sem_block_on_mask=0`). If the latest mask is older than `sem_mask_max_age_ms`, the semantic branch is disabled for that frame (fail-open to geometry). Previous tracks feed GeoDF-Adaptive. Confirmed semantic and geometric candidates are unioned, clipped by `geodf_max_reject_ratio` and a minimum feature count, then optionally enter backend optimization with residual weights w_i.

a: Design principles.

- No oracle leakage: the estimator never reads VIODE level, ground truth, ATE/RPE, or summary files.
- Fail-open semantics: missing/stale masks never stall VINS.
- Union under a budget: OR fusion cannot wipe the track set.
- Train/hold-out hygiene: policy thresholds are frozen from `city_day` before inspecting hold-out ATE.

B. GEOMETRIC BRANCH (GEODF-ADAPTIVE)

A temporal fundamental matrix is estimated from track correspondences. RANSAC inliers/outliers and Sampson residuals form a dual gate; tracks accumulate votes over `geodf_vote_frames`. Scene activation uses an EMA of the epipolar outlier ratio with hysteresis around an auto-calibrated ρ_{on}. Confirmed geometric candidates enter the shared reject set whenever GeoDF is armed.

C. SEMANTIC BRANCH

YOLO produces a dynamic-class mask on cam0. Per-track hits are vote-confirmed over `sem_vote_frames`. A semantic scene gate tracks an EMA of the dynamic pixel ratio with hysteresis. Soft masking of new features and hard cull of existing tracks are policy-dependent (Sec. III-E).

D. BIDIRECTIONAL OVERLAP ON RAW GEODF CANDIDATES

An earlier implementation computed overlap only on GeoDF confirmed tracks while `frame_active`, which empirically produced 0% overlap triggers even when both branches fired. Sem-GeoDF instead uses the raw GeoDF candidate pool G (fallback to confirmed if empty) and the semantic-raw set S (Fig. 4):

$$o = \max\left(\frac{|G \cap S|}{|G|}, \frac{|S \cap G|}{|S|}\right), \tag{1}$$

then applies EMA smoothing with coefficient `sem_policy_overlap_ema`. The candidate-count guard passes if $|G|$ or $|S|$ reaches `sem_policy_min_geo_candidates`. This keeps the overlap trigger alive when the two experts fire on differently sized track sets.

E. THREE-STATE ADAPTIVE POLICY

Let the discrete state be $s \in \{0, 1, 2\}$ (static-safe, dynamic-assist, strong-dynamic), as summarized in Fig. 5. Write r_t for the current-frame dynamic-pixel ratio, $\bar{r}_t$ for its EMA, and

TABLE 2. Key Sem-GeoDF parameters (defaults / train-selected).

Parameter	Value
`sem_vote_frames` / `geodf_vote_frames`	2 / 2
`sem_policy_burst_ratio`	0.18
`sem_policy_strong_ratio`	0.18 (selected)
`sem_policy_hold_frames`	180 (selected)
`sem_policy_overlap_ratio`	0.35
`geodf_max_reject_ratio`	0.40
`sem_mask_max_age_ms`	150
`sem_geodf_backend_weight`	1 (enabled)
`sem_geodf_backend_min_weight`	0.25
`sem_geodf_backend_recovery`	0.20

$\bar{o}_t$ for the overlap EMA of Sec. III-D. Escalation uses three online predicates and a hold timer of H frames:

$$\text{burst} = [r_t \geq \tau_{\text{burst}}], \tag{2}$$

$$\text{strong} = [\bar{r}_t \geq \tau_{\text{strong}}], \tag{3}$$

$$\text{overlap} = [\bar{o}_t \geq \tau_{\text{ov}} \ \wedge \ (|G| \vee |S|) \geq n_{\min}], \tag{4}$$

with defaults / train-selected values in Table 2. The hold timer prevents chatter after a burst or overlap trigger.

- $s{=}0$ (static-safe): soft-mask new features when the semantic scene is active; semantic hard cull off.
- $s{=}1$ (dynamic-assist): entered on burst, overlap, or an active hold; vote-confirmed semantic tracks enter the shared reject set when the semantic scene is active; soft-mask may be held for H frames.
- $s{=}2$ (strong-dynamic): entered on strong EMA and/or assist plus usable raw geo evidence; full OR fusion armed under the same shared reject budget.

Final selected values are written to `selected_params.yaml` from `city_day` replay only (never from hold-out ATE).

F. ADAPTIVE BACKEND RESIDUAL WEIGHTING

Hard rejection is not always possible: a shared ratio guard and minimum-feature constraints may keep suspicious tracks alive. When `sem_geodf_backend_weight=1`, each surviving track i carries a residual weight $w_i \in [w_{\min}, 1]$, broadcast to all of its observations and combined as the per-pair minimum across the two frames of a factor (Fig. 6). Every visual factor in Ceres multiplies both the residual and all of its Jacobian blocks (including the time-offset term) by $\sqrt{w_i}$, so the contribution is a genuine weighted least-squares term rather than an untracked post-hoc score. The weight is thus a per-track confidence, not an independent per-observation noise model; we state this explicitly to avoid overclaiming.

From online evidence at the current frame we form three risk terms in $[0, 1]$:

$$\rho_i^{\text{sem}} = \nVdash_i^{\text{sem}} (1 - w_{\text{sem}})\, c_i^{\text{sem}}, \tag{5}$$

$$\rho_i^{\text{geo}} = \nVdash_i^{\text{geo}} (1 - w_{\text{geo}})\, \max(c^{\text{geo}},\, e_i), \tag{6}$$

$$\rho_i^{\text{agr}} = \nVdash_i^{\text{sem}} \nVdash_i^{\text{geo}} (1 - w_{\text{agr}})\, \max(\bar{o}_t,\, \min(c_i^{\text{sem}}, \max(c^{\text{geo}}, e_i))), \tag{7}$$

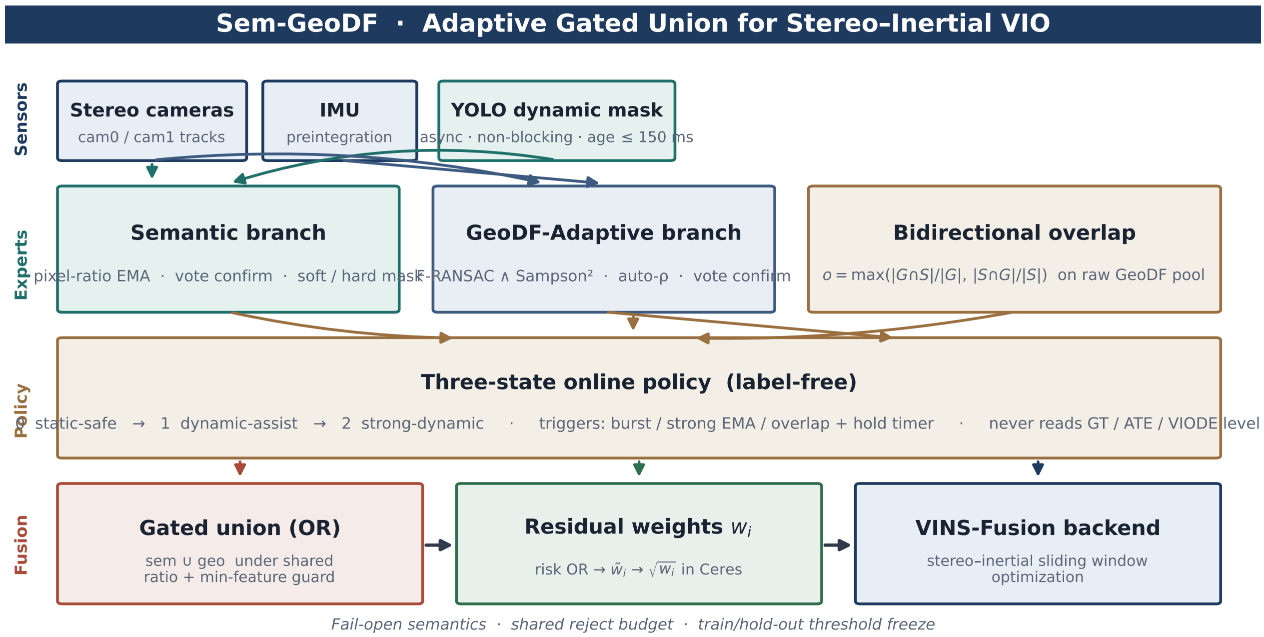

FIGURE 2. Sem-GeoDF system overview. Sensors feed a semantic expert (YOLO) and a GeoDF-Adaptive geometric expert; bidirectional overlap and a label-free three-state p

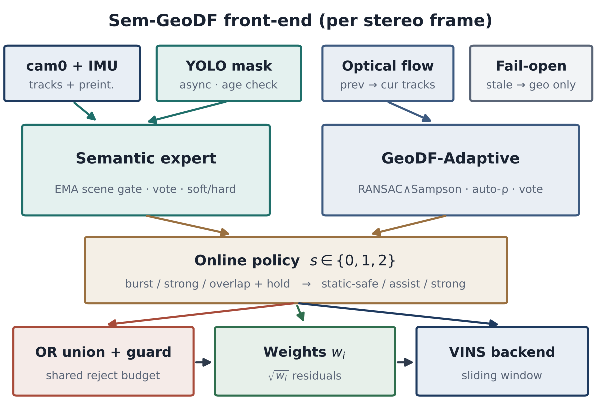

FIGURE 3. Per-frame Sem-GeoDF front-end: fail-open mask sync, dual experts, online policy, gated OR under a shared reject budget, and $\sqrt{\tilde{w}_i}$ residual weighting into VINS.

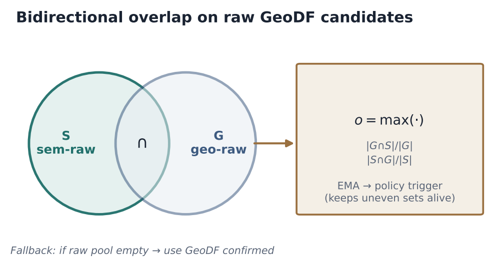

FIGURE 4. Bidirectional overlap on the raw GeoDF pool. Taking the maximum of the two c

where $\mathbb{1}_i^{\text{sem}}$ / $\mathbb{1}_i^{\text{geo}}$ flag semantic / GeoDF raw-or-confirmed hits, c^{sem} / c^{geo} are scene confidences from EMA+hysteresis (and policy state), and e_i is the normalized Sampson excess. A probabilistic-OR combines them into a target weight

$$\tilde{w}_i = \text{clip}\big(1-(1-\rho_i^{\text{sem}})(1-\rho_i^{\text{geo}})(1-\rho_i^{\text{agr}}),\ w_{\min},\ 1\big), \tag{8}$$

then caps $\tilde{w}_i$ by w_{sem}, w_{geo}, or w_{agr} when the corresponding confirmation flags fire. Per-id recovery with rate α_{rec} raises w_i toward a higher target gradually so transient false positives do not permanently suppress a track:

$$w_i \leftarrow \begin{cases} w_i^{\text{prev}} + \alpha_{\text{rec}}(\tilde{w}_i - w_i^{\text{prev}}) & \text{if } \tilde{w}_i > w_i^{\text{prev}}, \\ \tilde{w}_i & \text{otherwise.} \end{cases} \tag{9}$$

Audit columns in `sem_geodf_stats.csv` record `weighted_tracks`, `mean_backend_weight`, and related diagnostics. Table 7 lists the defaults used in paper configs.

IV. EXPERIMENTAL SETUP

A. IMPLEMENTATION

We use a ROS 2 Humble stereo-inertial VINS derived from VINS-Fusion [2]. YOLO runs on CUDA. All methods share a fair bag playback rate of $1.0\times$ (including non-YOLO baselines), so semantic methods are not artificially slowed relative to geometry-only runs.

B. DATASETS

VIODE [3]: environments `city_day`, `city_night`, and `parking_lot`, each with dynamic levels `0_none`,

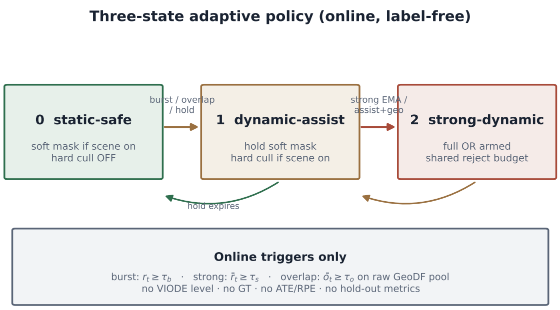

FIGURE 5. Three-state online policy. Escalation uses burst, strong semantic EMA, and bidirectional overlap only; the estimator never reads VIODE level, ground truth, or trajec

Algorithm 1 Sem-GeoDF front-end step (per stereo frame)

1: Sync latest YOLO mask if age $\leq$ max age; else disable semantic branch
2: Update semantic EMA; run GeoDF-Adaptive $\rightarrow$ raw/confirmed sets
3: Compute bidirectional overlap o on raw GeoDF pool; update policy s
4: $R \leftarrow$ GeoDF confirmed $\cup$ (policy-gated semantic confirmed)
5: Apply shared ratio / min-feature guard to R; soft-mask if allowed
6: If backend weighting enabled, assign w_i for survivors and recover idle ids

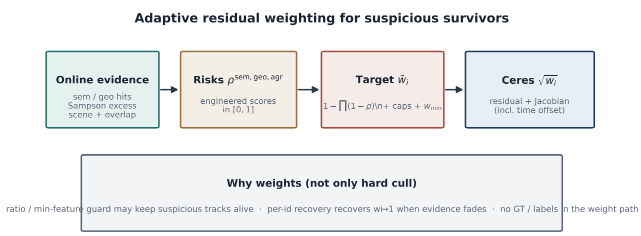

FIGURE 6. Adaptive residual-weight path for tracks that survive the shared reject budget. Online semantic/geometric risks form a probabilistic-OR target $\tilde{w}_i$, recovered per id, t

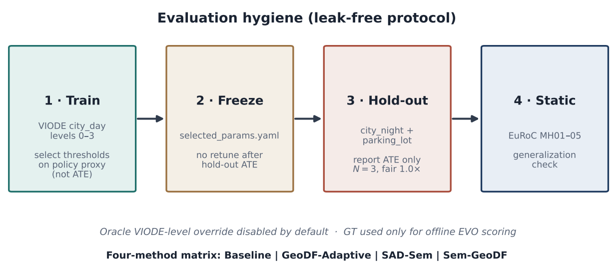

FIGURE 7. Leak-free evaluation protocol: thresholds are selected on VIODE `city_day` o

`1_low`, `2_mid`, `3_high`. **EuRoC** [17]: Machine Hall sequences MH_01–MH_05 for static generalization.

C. METHODS

Four methods are compared via `METHODS`:

- **Baseline:** stereo-inertial VINS without dynamic rejection.
- **GeoDF-Adaptive:** geometry-only adaptive rejection.
- **SAD-Sem:** semantic-only YOLO masking.
- **Sem-GeoDF (ours):** adaptive gated union + backend weights.

Optional sequential / mask-gated ablations exist in code but are excluded from the default matrix to keep the comparison focused.

D. PROTOCOL

Policy thresholds are selected on `city_day` only by maximizing a weighted static/dynamic policy objective on logged stats (not ATE). Hold-out night/parking ATE and EuRoC are reported after freezing thresholds (Fig. 7). Primary metric: ATE RMSE after SE(3) Umeyama alignment (EVO). On VIODE we run $N = 3$ trials per cell after an $N{=}1$ gate used to catch diverged runs; EuRoC uses $N = 1$–2 trials per cell (fewer than VIODE), so we treat it as a static generalization check rather than a full variance estimate. Every table cell prints its own trial count as a superscript, so $N{=}1$ cells are never mistaken for averaged means. All tables, figures, and abstract numbers are emitted by a single script (`make_sem_geodf_paper_assets.py`) directly from the run artifacts, so the manuscript is reproducible from the result tree.

V. RESULTS

A. VIODE TRAJECTORY ACCURACY

Fig. 8 summarizes Sem-GeoDF versus GeoDF-Adaptive on VIODE (negative = better). The largest gains concentrate on high-dynamic hold-out night and parking scenes, where geometry alone is insufficient. As Table 4 separates, on the eight dynamic hold-out cells Sem-GeoDF is better on 6, worse on 2, with the two regressions at the low-dynamic `0_none`/`1_low` end where semantics add little. We therefore claim spectrum consistency on hard scenes, not universal per-cell dominance.

B. EUROC STATIC GENERALIZATION

Fig. 9 highlights selected static and high-dynamic scenes across the four methods.

C. INTERPRETATION VERSUS SAD-SEM

The large negative deltas in Table 4 are measured against GeoDF-Adaptive, i.e. the geometry-only branch, which is the weaker baseline on high-dynamic night/parking scenes. Against semantic-only SAD-Sem the picture is different: on those same hard cells SAD-Sem often achieves the best per-cell ATE (e.g. `city_night_3_high` SAD-Sem ≈ 0.273 vs ours $\approx$ 0.354), and the mean ATE of Sem-GeoDF is typically close to SAD-Sem rather than strictly better. We

TABLE 3. VIODE ATE RMSE [m] (mean±std). Superscript is the number of trials N per cell; cells with $N{=}1$ report a single run (no std). Best mean per row in bold.

Scene	Baseline	GeoDF-Adaptive	SAD-Sem	Sem-GeoDF (ours)
city_day_0_none	$\mathbf{0.110{\pm}0.000}^3$	$0.120{\pm}0.000^3$	$0.150{\pm}0.012^3$	$0.116{\pm}0.000^3$
city_day_1_low	$0.139{\pm}0.001^3$	$0.155{\pm}0.000^3$	$\mathbf{0.137{\pm}0.000}^3$	$0.154{\pm}0.002^3$
city_day_2_mid	$0.166{\pm}0.000^3$	$\mathbf{0.142{\pm}0.000}^3$	$0.148{\pm}0.001^3$	$0.159{\pm}0.000^3$
city_day_3_high	$0.345{\pm}0.002^3$	$0.297{\pm}0.020^3$	$\mathbf{0.162{\pm}0.019}^3$	$0.245{\pm}0.020^3$
city_night_0_none	$0.418{\pm}0.000^3$	$0.307{\pm}0.107^3$	$\mathbf{0.298{\pm}0.005}^3$	$0.342{\pm}0.000^3$
city_night_1_low	$0.500{\pm}0.000^3$	$0.568{\pm}0.000^3$	$\mathbf{0.357{\pm}0.073}^3$	$0.452{\pm}0.000^3$
city_night_2_mid	$0.497{\pm}0.000^3$	$0.460{\pm}0.000^3$	$0.372{\pm}0.000^3$	$\mathbf{0.281{\pm}0.013}^3$
city_night_3_high	$0.883{\pm}0.001^3$	$0.822{\pm}0.000^3$	$\mathbf{0.273{\pm}0.030}^3$	$0.354{\pm}0.000^3$
parking_lot_0_none	$0.168{\pm}0.001^3$	$0.149{\pm}0.000^3$	$0.188{\pm}0.000^3$	$\mathbf{0.139{\pm}0.000}^3$
parking_lot_1_low	$0.107{\pm}0.018^3$	$\mathbf{0.106{\pm}0.000}^3$	$0.132{\pm}0.000^3$	$0.113{\pm}0.006^3$
parking_lot_2_mid	$0.144{\pm}0.000^3$	$0.205{\pm}0.000^3$	$\mathbf{0.088{\pm}0.000}^3$	$0.099{\pm}0.000^3$
parking_lot_3_high	$0.119{\pm}0.000^3$	$0.172{\pm}0.000^3$	$0.146{\pm}0.003^3$	$\mathbf{0.106{\pm}0.037}^3$

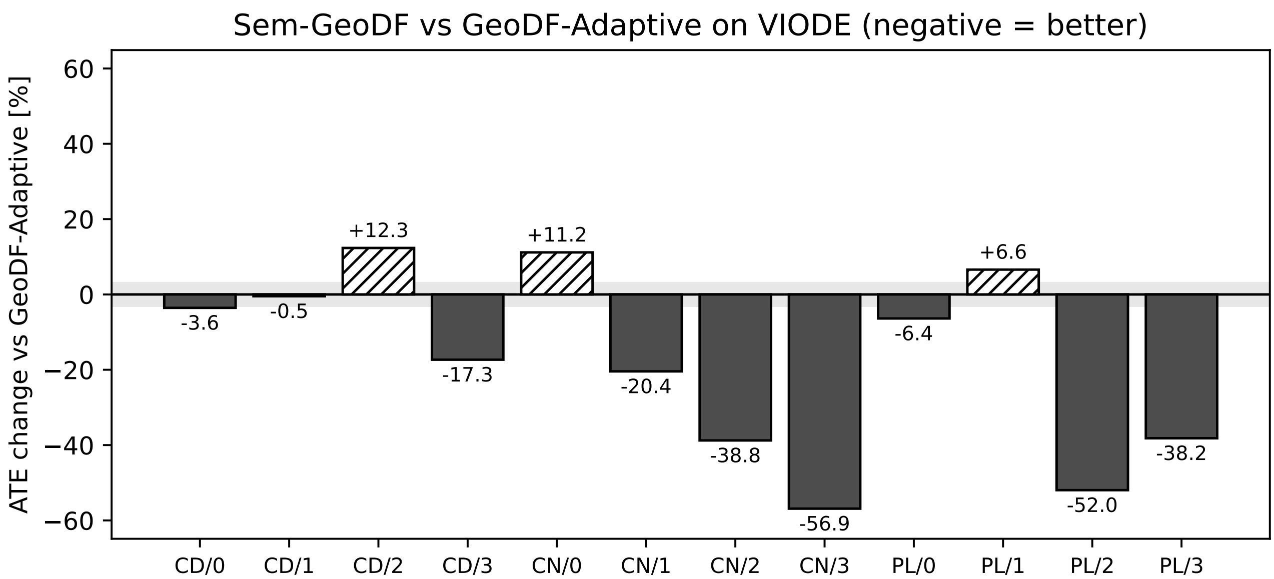

FIGURE 8. ATE change of Sem-GeoDF versus GeoDF-Adaptive on VIODE (negative = better). Shaded band: ±3%.

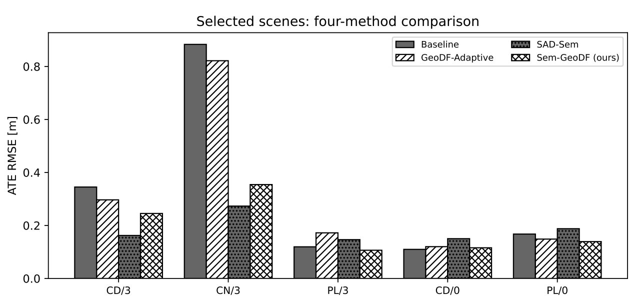

FIGURE 9. Selected static and high-dynamic scenes: four-method ATE comparison.

therefore state the headline explicitly as gains over geometry-only filtering, and we do not market Sem-GeoDF as a strict ATE winner over SAD-Sem. Its value for IEEE Access is the combination of: (i) large gains over geometry-only filtering on hard hold-outs; (ii) reduced static regression relative to always-on semantics (Sec. V-D); (iii) label-free online control, shared reject budget, and adaptive residual weights; (iv) a reproducible fair-rate protocol whose tables regenerate exactly from the run artifacts.

D. STATIC SAFETY

On `0_none` levels, always-on semantics can regress relative to Baseline because false-positive masks remove useful structure. Table 6 quantifies the effect: on `city_day_0_none`, SAD-Sem is $+36\%$ versus Baseline while Sem-GeoDF stays within $+5\%$; on `parking_lot_0_none`, SAD-Sem is $+12\%$ while Sem-GeoDF is -17% (best of the four). `city_night_0_none` is mixed—both Adaptive and SAD-Sem improve over Baseline, and Sem-GeoDF sits between them—but the static-safe policy still avoids the day/parking over-rejection pattern of always-on masking. This is the practical justification for a policy rather than

TABLE 4. Sem-GeoDF ATE change versus GeoDF-Adaptive (negative = better; both means over the same N). Decision band $\pm 3\%$. Train (`city_day`) is separated from hold-out; EuRoC is a static generalization check.

Scene	Δ vs Adaptive [%]	Verdict	N
Train: city_day (thresholds selected here)			
city_day_0_none	-3.6	better	3
city_day_1_low	-0.5	similar	3
city_day_2_mid	+12.3	worse	3
city_day_3_high	-17.3	better	3
subtotal: 2 better / 1 similar / 1 worse			
Hold-out: city_night, parking_lot			
city_night_0_none	+11.2	worse	3
city_night_1_low	-20.4	better	3
city_night_2_mid	-38.8	better	3
city_night_3_high	-56.9	better	3
parking_lot_0_none	-6.4	better	3
parking_lot_1_low	+6.6	worse	3
parking_lot_2_mid	-52.0	better	3
parking_lot_3_high	-38.2	better	3
subtotal: 6 better / 0 similar / 2 worse			
EuRoC (static generalization)			
MH_01_easy	+3.7	worse	2
MH_02_easy	-8.0	better	2
MH_03_medium	-0.0	similar	1
MH_04_difficult	-2.2	similar	1
MH_05_difficult	-1.7	similar	1
subtotal: 1 better / 3 similar / 1 worse			
Hold-out headline: 6 better / 0 similar / 2 worse on the 8 dynamic hold-out scenes.			

always-on semantics.

E. ROLE OF ADAPTIVE RESIDUAL WEIGHTS

Backend weighting complements hard cull: when the ratio guard preserves suspicious tracks, down-weighting reduces their influence without emptying the feature set. Because weights are recovered per id, transient false positives do not permanently suppress a track. This mechanism is active in Sem-GeoDF paper configs (`sem_geodf_backend_weight`=1) and is absent from Baseline / Adaptive / SAD-Sem.

F. BACKEND WEIGHT MECHANISM

Table 7 summarizes the residual-weight parameters used throughout the Sem-GeoDF paper runs. A paired ATE ablation that disables only w_i (`sem_geodf_backend_weight`=0) while freezing the identical gated union and train-selected policy is reserved for camera-ready once the `sem_geodf_noweight` matrix finishes; until then we do not invent numerical deltas. The equations in Sec. III-F already make the contribution of w_i explicit relative to hard cull under the shared reject budget.

G. SENSITIVITY AND REALTIME BEHAVIOR

Offline replay on `city_day` shows that nearby (`burst`, `strong`, `hold`) grid points yield similar weighted policy objectives. Overlap thresholds 0.30–0.40 change little when overlap events are sparse—consistent with burst/strong dominating escalation on many sequences. Non-blocking mask sync keeps VINS realtime-capable: the estimator never waits on YOLO; it either consumes a fresh-enough mask or falls back to GeoDF-only. Logged `sem_mask_lag_ms` provides an audit of mask age at consumption time. The $\pm 3\%$ decision band is a reporting convention (roughly the run-to-run ATE spread we observe on repeated VIODE trials), used only to label a cell better/similar/worse; it is not a significance test, and all raw deltas are shown so readers can apply their own threshold.

H. FAILURE MODES AND LIMITATIONS

1) Mid-dynamic day cells where GeoDF already helps and adding semantics can slightly increase ATE; two low-dynamic hold-out cells (`city_night_0_none`, `parking_lot_1_low`) also regress versus GeoDF-Adaptive.
2) The three policy escalation triggers (burst, strong, overlap) are all driven by semantic signals; geometry never escalates the semantic branch on its own. Sem-GeoDF is thus best described as "geometry always-on with its own gate, OR policy-gated semantics," not a symmetric consensus.
3) The residual-weight mechanism is specified and enabled (Sec. III-F, Table 7), but a paired ATE ablation with w_i off is still pending camera-ready completion of the `sem_geodf_noweight` matrix; we therefore do not claim a quantified ATE split between union and weighting yet.
4) EuRoC uses N = 1–2 trials per cell (fewer than VIODE); it is a static generalization check, not a full variance estimate. $N = 3$ on VIODE is also small for formal significance testing.
5) VIODE is simulated; real-world dynamic bags remain future work.
6) Matched external SOTA tables (e.g., DynaVINS under identical bags/rates) are not included here.

VI. DISCUSSION

IEEE Access is the target venue because the contribution is a reproducible applied fusion module with honest trade-offs, rather than a short letter claiming frontier novelty over full dynamic-VIO stacks. The manuscript follows the official Access two-column layout (`ieeeaccess.cls`).

a: Claim framing.

We claim spectrum robustness and static safety: large gains versus GeoDF-Adaptive on high-dynamic hold-outs (Table 4), reduced static regression versus always-on semantics (Table 6), and a label-free online policy under a fair $1.0\times$ protocol. We do not claim universal per-cell ATE superiority over SAD-Sem, matched DynaVINS SOTA under identical bags, or a finished paired w_i ATE ablation.

b: Reproducibility.

Each run writes `run_manifest.json` (git SHA, config hash, bag rate, oracle flag) and per-frame `sem_geodf_stats.csv`. Policy thresholds can be replayed offline with `simulate_sem_policy.py`. The

TABLE 5. EuRoC Machine Hall ATE RMSE [m]. Superscript is the number of trials N per cell; cells with $N{=}1$ report a single run (no std). Best mean per row in bold.

Scene	Baseline	GeoDF-Adaptive	SAD-Sem	Sem-GeoDF (ours)
MH_01_easy	$\mathbf{0.166{\pm}0.021}^2$	$0.177{\pm}0.000^2$	$0.175{\pm}0.000^2$	$0.183{\pm}0.000^2$
MH_02_easy	$0.169{\pm}0.000^2$	$0.169{\pm}0.000^2$	$0.158{\pm}0.000^2$	$\mathbf{0.155{\pm}0.000}^2$
MH_03_medium	0.292^1	0.274^1	$\mathbf{0.261}^1$	0.274^1
MH_04_difficult	0.447^1	0.441^1	0.440^1	$\mathbf{0.432}^1$
MH_05_difficult	0.298^1	0.298^1	0.294^1	$\mathbf{0.293}^1$

TABLE 6. Static-scene ATE RMSE [m] on VIODE `0_none` ($N{=}3$). Δ vs Baseline in parentheses (negative = better).

Scene	Base.	Adapt.	SAD-Sem	Sem-GeoDF
city_day_0_none	0.110	0.120 (+9%)	0.150 (+36%)	**0.116** (+5%)
city_night_0_none	0.418	0.307 (−27%)	**0.298** (−29%)	0.342 (−18%)
parking_lot_0_none	0.168	0.149 (−11%)	0.188 (+12%)	**0.139** (−17%)

TABLE 7. Adaptive residual-weight mechanism (enabled in Sem-GeoDF paper configs; `sem_geodf_backend_weight`=1). A paired ATE ablation with w_i off under an identical gated union is planned for camera-ready once `sem_geodf_noweight` runs finish; Baseline / Adaptive / SAD-Sem do not use this term.

Symbol / flag	Default	Role
$w_{\min}$	0.25	Floor on residual weight
w_{sem}	0.55	Cap when semantic-confirmed
w_{geo}	0.75	Cap when GeoDF-confirmed
w_{agr}	0.25	Cap when both confirmed
α_{rec}	0.20	Per-id recovery toward target
Ceres scale	$\sqrt{w_i}$	Residual and full Jacobian

`paper_postfix` result tree is generated with the corrected $\sqrt{w_i}$ scaling on the time-offset Jacobian (Sec. III-F), so the released estimator reproduces the reported Sem-GeoDF numbers without a residual/Jacobian mismatch. The default paper matrix is launched by

```
METHODS="baseline adaptive sad_sem sem_geodf".
```

Code, configs, and evaluation scripts will be released upon acceptance.

VII. CONCLUSION

Sem-GeoDF couples semantic and geometric dynamic-feature rejection through an online adaptive gated union for stereo-inertial VIO, with optional adaptive backend residual weighting for suspicious survivors. With label-free control and a strict train/hold-out protocol, it improves several high-dynamic VIODE hold-outs over geometry-only filtering while mitigating semantic over-rejection in static scenes. We present this work as an engineering contribution for IEEE Access with full tables, figures, explicit limitations, and a clear path to camera-ready weight ablation.

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